

Adish Yermal

+91-7795772301 — adishy.is19@gmail.com — LinkedIn — Portfolio

PROFESSIONAL SUMMARY

Software Development Engineer with strong expertise in Embedded Systems, Backend Engineering, and QA Automation. Experienced in building scalable distributed systems using C++, Python, Qt/QML, GTK, and PHP (Laravel). Strong background in Linux systems programming, multithreading, IPC, ZeroMQ-based architectures, REST APIs, CI/CD pipelines, and test automation frameworks. Proven experience across Healthcare, Banking, Embedded Devices, and Enterprise applications.

TECHNICAL SKILLS

Programming: C, C++, Python, PHP, SQL, JavaScript

Embedded & Systems: Linux, Embedded Systems, Multithreading, IPC, Yocto, Device Drivers, System Programming

Backend & Distributed: REST APIs, ZeroMQ, Microservices Concepts, Backend Architecture

QA / Automation: Selenium concepts, API Testing, Postman, Test Validation, Debug Automation, Regression Testing

Frameworks: Qt/QML, GTK, Laravel, Firebase

Tools: Git, CMake, GTest, JIRA, Jenkins (CI/CD), Postman

Databases: MySQL, Google Cloud Platform

PROFESSIONAL EXPERIENCE

Software Development Engineer – Tata Elxsi, Bangalore

Oct 2023 – Present

Project: Embedded Healthcare Systems — Domain: Medical Devices / Embedded / SDE

- Developed embedded healthcare applications using C++ and Qt/QML on Linux-based systems
- Built distributed communication system using ZeroMQ with multithreaded architecture
- Designed middleware for logging, encryption, system communication, and database integration
- Developed Python-based automation tools for debugging, validation and Application Packaging.
- Built multilingual translation framework improving UI scalability and maintainability
- Developed CLI-based tools for system benchmarking, validation, and performance analysis
- Worked in Agile/Scrum environment with CI/CD-based delivery pipelines

CLIENT PROJECTS

Skannay Technologies – Embedded Software Engineer

Domain: Medical Embedded Systems — Linux / IPC / Device Software

- Developed GTK-based embedded applications on Linux systems for medical devices
- Designed ZeroMQ-based distributed messaging architecture for IPC communication
- Implemented AES encryption modules for secure medical data transmission
- Built multithreaded benchmarking and performance analysis tools
- Developed TCP/IP and IPC communication layers for embedded systems
- Built system-level visualization tools for real-time performance monitoring
- Contributed to Yocto-based embedded Linux firmware development

Neogen 3M Clean Trace – Qt/QML Engineer

Domain: Healthcare Device Software — Embedded UI Systems

- Developed high-performance UI screens using Qt/QML for medical diagnostic systems
- Improved UI rendering efficiency by 25% using reusable component architecture
- Designed multilingual localization framework integrated with proprietary translation engine
- Developed REST API middleware for frontend-backend communication
- Performed API validation and testing using Postman (QA involvement)
- Integrated Linux hardware-level services including MAC ID and system device APIs

INTERNAL PROJECTS

LMS Platform – Full Stack Developer (Laravel / PHP)

- Developed a role-based learning management system using PHP Laravel and MySQL
- Built backend APIs for workflow automation and user management
- Improved system efficiency by reducing manual administrative operations
- Collaborated with UI and embedded teams for system integration

INTERNSHIP

Software Intern – Kashika Agritech Pvt Ltd

- Improved SCM/CMS workflows and system optimization
- Supported feature development and debugging activities

Cybersecurity Intern – Center for Excellence in Digital Forensic investigation and cyber security

- Led Network team, set up network server for attack simulation.
- Conducted a 3 day Flagship workshop with cyber sapiens
- Attended a workshop with the CyberCrime police department in Mangalore.

PROJECTS

HR ops (Node.js, Electron, Tailwind, HTML, CSS, JavaScript)

- Developed HR based Employee Management platform for Employee life cycle
- SKU, Inventory management and Invoice generation features

Railway Ticket Management System (PHP + MySQL)

- Developed a role-based booking system with database integration

E-Waste Management System (Firebase, Java, Android)

- Built real-time waste collection and trading platform

LEADERSHIP & ACTIVITIES

- Alumni Secretary - Sahyadri College of Engineering and management (2023 -2024)
- Placement President – Sahyadri College of Engineering management (2023)
- IEEE Student Chair - Sahyadri College of Engineering and management (2022)
- Conducted cybersecurity and technical workshops
- Toycathon 2021 Finalist: His project, "Indus Trail," was selected for the Grand Finale of this national-level competition organized by the Ministry of Education
- Published Author: Co-authored research on OAuth 2.0 security defenses using digital signatures, featured in the book Advancements in Cybercrime Investigation and Digital Forensics

EDUCATION

Bachelor of Engineering – Information Science

2019 – 2023

Sahyadri College of Engineering and Management, Mangalore

Relevant Focus Areas: Software Engineering, Data Structures, Operating Systems, Computer Networks, Embedded Systems

CERTIFICATIONS

- Tata Elxsi Ray of Hope Certificate
- Google IT Support Professional Certificate
- Google Cloud Fundamentals
- Google Cloud Identity
- AWS Machine Learning Fundamentals